

MAKE UP EXAMINATIONS – JUNE 2023

Program	: B.E :- Common to CSE / ISE / AI & ML / AI & DS	Semester	: III
Course Name	: Linear Algebra and Laplace Transforms	Max. Marks	: 100
Course Code	: CS/IS/AI/AD31	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Define a periodic function with an example. CO1 (06)
 - Evaluate $L\left\{\frac{\cos 4t - \cos 2t}{t}\right\}$.
 - If $L\{f(t)\} = F(s)$, then prove that $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}\{F(s)\}$, where n is a positive integer. CO1 (07)
 - Given $L\{2\sin\sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$, show that $L\left\{\frac{\cos\sqrt{t}}{\sqrt{t}}\right\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$. CO1 (07)
- Find the Laplace transform of $e^{-2t}(2\cos 5t \sin 3t)$. CO1 (06)
 - Evaluate $L\left\{\int_0^t t e^{-t} \cos t dt\right\}$
 - If $L\{f(t)\} = F(s)$, then prove that $L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s) ds$ and hence find $L\left\{\frac{\sin 4t}{t}\right\}$. CO1 (07)
 - Find the Laplace transform of the triangular wave function of period $2a$ is given by $f(t) = \begin{cases} t & \text{for } 0 \leq t \leq a \\ 2a - t & \text{for } a \leq t \leq 2a \end{cases}$. CO1 (07)

UNIT - II

- Find (i) $L^{-1}\left\{\frac{e^{-s}(s+2)}{(s+1)^2}\right\}$ (ii) $L^{-1}\left\{\tan^{-1}\left(\frac{2}{s}\right)\right\}$. CO2 (06)
 - Using convolution theorem find $L^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}$. CO2 (07)
 - Solve $y'' + 4y' + 3y = e^{-t}$; given $y(0) = 0$, $y'(0) = 0$ by the method of Laplace transform. CO2 (07)

4. a) Find the inverse Laplace transform of $\frac{5s+3}{(s-1)(s^2+2s+5)}$. CO2 (06)
- b) Express $f(t) = \begin{cases} \cos t, & 0 < t < \pi, \\ \cos 2t, & \pi < t < 2\pi, \\ \cos 3t, & t > 2\pi. \end{cases}$ in terms of the Heaviside function CO2 (07)
- and hence find its Laplace transform.
- c) The coordinates (x, y) of a particle moving along a plane curve at any point t , are given by $y' + 2x = \sin 2t$; $x' - 2y = \cos 2t$; $t > 0$. If at $t = 0$, $x = 1$ and $y = 0$ show by using Laplace Transforms that the particle moves along the curve $4x^2 + 4xy + 5y^2 = 4$. CO2 (07)

UNIT - III

5. a) Determine whether the vector $(1, -2)$ is a linear combination of the vectors $(2, 4)$ and $(3, 6)$. CO3 (06)
- b) Prove that the transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(x, y) = (2x, x + y)$ is linear. Find the image of $(1, 2)$ and $(-1, -4)$ under this transformation. CO3 (07)
- c) Consider the bases $B = \{(1, 2), (3, -1)\}$ and $B' = \{(1, 0), (0, 1)\}$ of $\mathbb{R}^2$. Find the transition matrix from B to B' . If u is a vector such that $U_B = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, find $U_{B'}$. CO3 (07)
6. a) Define linear dependence and independence of vectors check the linear independence of a set of vectors $\{(1, 0, 1), (2, 1, 3), (-1, 3, 2)\}$. CO3 (06)
- b) Define kernel and range of a linear transformation. Verify Rank nullity theorem for the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$. CO3 (07)
- c) Determine the matrix that describes a reflection in the x -axis, followed by a rotation through $\frac{\pi}{2}$, followed by contraction of factor $\frac{1}{2}$. Find the image of $\begin{bmatrix} 4 \\ -1 \end{bmatrix}$ under this sequence of mappings. CO3 (07)

UNIT- IV

7. a) Find a basis for $\text{col}(A)$ and $\text{nul}(A)$ of $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$. CO4 (10)
- b) Find the complete solution of $x_1 + 2x_2 + 3x_3 + 5x_4 = b_1$, $2x_1 + 4x_2 + 8x_3 + 12x_4 = b_2$, $3x_1 + 6x_2 + 7x_3 + 13x_4 = b_3$ by choosing appropriate B . CO4 (10)
8. a) Find QR factorization of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. CO4 (10)
- b) Find a least-squares solution of $Ax = b$ where $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$. CO4 (10)

UNIT - V

9. a) Orthogonally diagonalize the matrix $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$. CO5 (10)
- b) (i) Determine whether the matrix $A = \begin{bmatrix} 0 & 2+i & 1 \\ 2-i & i & 0 \\ 1 & 0 & 1 \end{bmatrix}$ is Hermitian or not. CO5 (10)
- (ii) Check whether the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ is positive definite or not.
10. a) (i) Diagonalize the matrix $A = \begin{bmatrix} -1 & 4 \\ 0 & 3 \end{bmatrix}$ and hence find A^5 . CO5 (10)
- (i) Determine whether the matrix $A = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$ is Unitary or not.
- b) Find a singular value decomposition of $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$. CO5 (10)
